

Requests to wolfSSL

Lightrider Inc, Veloce PQC SDK. Certificate #4718, module v5.2.1, bundle wolfssl-5.9.2-commercial-fips-linuxv5.2.1, Agreement v.01-24.

1 Our configuration, for your verification

Linux build	<code>./configure --enable-fips=v5</code> ; make, fips-hash.sh, make, testwolfcrypt. No <code>--enable-wolfEntropy</code> , per your 2026-08-27 guidance that wolfEntropy was never tested with v5.2.1.
Windows build	UG 2.2.45 (OE49): wolfssl-fips.sln, DLL Release x64, Visual Studio 2026, no PAA, your supplied <code>user_settings.h</code> (HAVE_FIPS_VERSION 5 / MINOR 2 / PATCH 1).
Seeding	Custom generate-seed callback registered with <code>wc_SetSeed_Cb</code> (FIPS FAQ §2). Revised 2026-09-17: the callback reads the processor's RDSEED instruction (conditioned output of the CPU's SP 800-90B noise source) and delivers it directly to the module's Hash_DRBG, with SP 800-90B RCT (cutoff 4) and APT (13/512) health tests at $H = 8$ bits/sample and a 1024-sample startup test; fails closed on any error or when RDSEED is absent. Operating system DRBG output (<code>getrandom</code> , <code>BCryptGenRandom</code>) is no longer the default: it is an SP 800-90C RBGC chain and is available only by explicit configuration, reported with no security-strength claim. Documented under legacy IG 9.3.A; no ESV certificate claimed; caveat "no assurance of the minimum strength of generated SSPs" acknowledged.
Distribution	wolfCrypt as object code only inside Veloce installers; notices retained; source confined to a restricted private repository.

2 Please verify or confirm

1. The Linux and Windows build configurations above are approved configurations for module v5.2.1 under the User Guide / COGM.
2. The revised seeding design above (RDSEED direct to the module DRBG) raises no concern for v5.2.1 under legacy IG 9.3.A, and the earlier operating-system-DRBG seed source is correctly understood as an SP 800-90C RBGC construction that should not be the shipped default.
3. Authenticode signing of the FIPS DLL or of the MSI installer does not interfere with the in-core HMAC-SHA-256 integrity check.

3 Please notify us

1. When **OE49** (Windows 11 Pro, Lenovo ThinkPad T14, Intel i7-1260P) and **OE36** (Ubuntu 24.04 on Docker 28.5 / Hyper-V, Xeon Platinum 8272CL, Azure Standard_D2s_v4; CAVP validation 36918) appear on the published #4718 certificate and security policy, with any expected CMVP timeframe. Until publication we ship Windows with validation status "pending publication".

4 Please advise on entropy source validation

Your review noted that `getrandom` and `BCryptGenRandom` return the output of the operating system's SP 800-90A DRBG, so seeding the module DRBG from them forms an SP 800-90C RBGC construction whose randomness source is not validated on our operational environments, and recommended seeding from raw SP 800-90B entropy instead. We have moved the default seed source to RDSEED accordingly. Under IG 9.3.A Additional Comment 11 an ESV certificate counts only when its number appears on the module certificate and in the security policy, so any ESV credit for Veloce depends on the module side. Please advise on the following.

1. **RDSEED as the external seed source.** For module v5.2.1 under legacy IG 9.3.A, is a generate-seed callback that delivers RDSEED output directly to the Hash_DRBG the seed design you recommend? Should the callback apply SP 800-90B RCT/APT health tests (as ours does), condition the output further (for example through a vetted conditioning function with oversampling), or deliver it unmodified? Which entropy estimate per 8-bit sample should the health-test cutoffs assume for Intel and AMD RDSEED?
2. **RDRAND and TPM.** We read RDRAND and TPM2_GetRandom as DRBG outputs, so both would again form RBGC chains, with a FIPS-validated TPM being a chain from a validated RBG. Please confirm or correct, and state whether a validated TPM is an acceptable alternative source where RDSEED is unavailable (arm64, some virtual machines).
3. **Windows CNG.** On Windows 11, `BCryptGenRandom` is served by Microsoft's FIPS-validated Cryptographic Primitives Library. Is an RBGC construction from that validated RBG acceptable for the module on OE49, and what documentation would the security policy or user guide expect?

4. **Entropy assessment path.** Whether wolfSSL can point to, or obtain under a commercial engagement, an ESV certificate or an SP 800-90B entropy assessment covering RDSEED on the Windows 11 x86-64 (OE49) and Ubuntu x86-64 operational environments, and whether such a certificate would be cited in a wolfCrypt security policy so that IG 9.3.A Additional Comment 11 is satisfied.
5. **ESV coverage of future modules.** The ESV certificate and operating-environment list that will back the v5.2.4 module and the FIPS v7 “PQ-FS” module, and whether Windows 11 x86-64 and Ubuntu 24.04 x86-64 will be covered. ESV #E335 currently lists one OE (AEDS Software Distribution 2.0, Xeon E3-1275 v6) with reuse restricted to vendor.
6. **Third-party sources.** Whether wolfSSL will cite a third-party ESV certificate (a Lightrider-owned entropy source or an IDQ Quantis QRNG) in a wolfCrypt module security policy under a commercial engagement, or alternatively offer an OEM module validation for Lightrider.
7. **Resubmission effect.** Whether the in-process resubmission of #4718 for OE49 and OE36 changes the module’s legacy IG 9.3.A status or its entropy caveat.

5 Please paper on the Order Form

1. Veloce as the named Program under §3: a PQC SDK exposing cryptographic services through a local agent, not a linkable cryptography library.
2. Commercial coverage for our PQC provider: standalone compilation of `wc_mlkem.*`, `wc_mldsa.*`, and `sha3.c` from the public `v5.9.2-stable` tree (the commercial bundle strips SHAKE), shipped as object code only.
3. Budget and timing for the FIPS v7 “PQ-FS” upgrade (§10 treats it as an upgrade requiring a new Order Form).